

Supplementary file for “Robust Fault Diagnosis for Cyber-Physical Systems Under Sensor-Reading Modification Attacks via Dynamic Observation”

Wenli Duo, *Member, IEEE*, Shouguang Wang, *Senior Member, IEEE*, MengChu Zhou, *Fellow, IEEE*, and Dan You, *Member, IEEE*

This file provides a complete proof for Theorem 3 in manuscript “Robust Fault Diagnosis for Cyber-Physical Systems Under Sensor-Reading Modification Attacks via Dynamic Observation”.

Theorem 3: Given plant G with Σ_f , a set of vulnerable events $\Sigma_v \subseteq \Sigma_o$, and D , let G_t^a be the test automaton constructed from D under SM attacks using *ConstructGta*. G is robustly diagnosable w.r.t. Σ_f , Σ_v , and D iff G_t^a contains no NSCC.

Proof: We first clarify certain properties that are necessary for the proof. Let $X_N = X \times \{N\}$ and $X_Y = X \times \{Y\}$. For a path $v \in L(G_t^a)$, $P_G(v)$ denotes the actual string generated by G . Let $m(v) \in (\Theta \cup \Sigma_o)^*$ denote the modified run w.r.t. v . Specifically, $\forall s \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^*$, and $e \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))$,

$$m(se) = \begin{cases} m(s)\theta, & \text{if } e = \theta \in \Theta \\ m(s)\sigma', & \text{if } e = (\sigma, \sigma') \in (\Sigma_o \times \Sigma_o) \\ m(s)\sigma, & \text{if } e = \sigma \in \Theta' \\ m(s), & \text{if } e = \sigma \notin \Theta' \end{cases},$$

where Θ' is the last sensing decision in s . According to construct of G_t^a , we have $\eta_t^a(z_{0,i}^a, v) = (\tilde{\eta}_D(\tilde{z}_{0,D}, m(v)), \delta(x_{0,i}, P_G(v)))$. Moreover, there exists a run $w \in L(D)$ such that $P_G(v) \in P_\Theta^{-1}(w)$ and $m(v) \in M_\Theta(w)$. In addition, we have $\tilde{\eta}_D(\tilde{z}_{0,D}, m(v)) = risk \Leftrightarrow m(v) \notin L(D)$. Based on the transition function of D , $\forall w \in L(D)$, if $\eta_D(z_{0,D}, w) = z$ or (z, θ) , then $z = \{x_l \in X_l | \exists s \in P_\Theta^{-1}(w) \cap L(G), \delta(x_{0,l}, s) = x_l\}$. Since $s \in L_N \Leftrightarrow \delta(x_{0,l}, s) \in X_N$, which indicates that

$$z \cap X_N \neq \emptyset \Leftrightarrow P_\Theta^{-1}(w) \cap L_N \neq \emptyset. \quad (1)$$

We now prove the two directions.

($\Rightarrow$) By contradiction, assume that G_t^a contains an NSCC, which involves a decision state (z, θ, x_l) , where

$$x_l \in X_Y \text{ and } z \cap X_N \neq \emptyset. \quad (2)$$

Let $v \in L(G_t^a)$ and $\eta_t^a(z_{0,i}^a, v) = (z, \theta, x_l)$. There exists $v' \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^* \setminus \{\varepsilon\}$, such that $\eta_t^a((z, \theta, x_l), v') = (z, \theta, x_l)$. Based on transition rule of G_t^a , $|P_G(v')| > 0$. Let $v_n = v(v')^n$, where $n \in \{1, 2, \dots\}$. Since $x_l \in X_Y$, a fault has occurred. $P_G(v_n)$ can be written as $P_G(v_n) = st_n$, where $s \in \Psi(\Sigma_f)$ and t_n can be made arbitrarily large by increasing n . According to construction of D , there exists $w_n \in L(D)$ such that $st_n \in P_\Theta^{-1}(w_n)$ and $m(v_n) \in M_\Theta(w_n)$. As $\eta_t^a(z_{0,i}^a, v_n) \notin \{risk\} \times X_l$, we have $m(v_n) \in L(D)$. Based on (1) and (2), we have $P_\Theta^{-1}(m(v_n)) \cap L_N \neq \emptyset \Rightarrow [P_\Theta^{-1}(m(v_n)) \cap L(G)] \not\subseteq L_F$. Based on Definition 6, G is not robustly diagnosable, which causes contradiction.

($\Leftarrow$) By contradiction, suppose that G is not robustly diagnosable. By Definition 6, it indicates that $\forall k \in \mathbb{N}$, there exist $s \in \Psi(\Sigma_f)$, $t \in L(G)/s$ with $|t| \geq k$, $w \in L(D)$, and $u \in M_\Theta(w)$, such that $st \in P_\Theta^{-1}(w)$, $u \in L(D)$ and $[P_\Theta^{-1}(u) \cap L(G)] \not\subseteq L_F$. According to construction of G_t^a , $\exists v_{N_l} \in L(G_t^a)$, such that $s = P_G(v_f)$, $t = P_G(v_l)$, and $u = m(v_{N_l})$. Since $st \in L_F$ and $u \in L(D)$, $\delta(x_{0,l}, st) \in X_Y$ and $\eta_t^a(z_{0,i}^a, v_{N_l}) \notin \{risk\} \times X_l$. Let $\eta_t^a(z_{0,i}^a, v_{N_l}) = (z, x_l)$ or (z, θ, x_l) . By (1), we have $z \cap X_N \neq \emptyset$. Since G_t^a is a finite state automaton, any path in G_t^a contains at most $|Z_t^a|$ different states. As $|t|$ can be arbitrarily large, we have $|t| \geq |Z_t^a| \Rightarrow |v_l| \geq |Z_t^a|$. v_{N_l} ultimately leads to a nontrivial SCC in G_t^a , since this trajectory contains at least $|Z_t^a| + 1$ states. It implies that $\exists v_{l1}, v_{l2}, v_{l3} \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^*$ with $v_{l2} \neq \varepsilon$, and $(z', \theta', x_l') \in Z_t^a$, such that $v_l = v_{l1}v_{l2}v_{l3}$, $\eta_t^a(z_{0,i}^a, v_{N_l}) = (z', \theta', x_l')$, and $\eta_t^a((z', \theta', x_l'), v_{l2}) = (z', \theta', x_l')$. Accordingly, $\exists t_1, t_2, t_3 \in \Sigma^*$, such that $t = t_1t_2t_3$, $t_1 = P_G(v_{l1})$, $t_2 = P_G(v_{l2})$, and $t_3 = P_G(v_{l3})$. We have $x_l' = \delta(x_{0,l}, st_1t_2) \in X_Y$. We then show that $z' \cap X_N \neq \emptyset$. If $z' \cap X_N = \emptyset$, then $z' \subseteq X_Y$ and $\eta_t^a((z', \theta', x_l'), v_{l3}) = (z, x_l)$ or (z, θ, x_l) , where $z \subseteq X_Y$. Clearly, it contradicts the fact that $z \cap X_N \neq \emptyset$. Thus, $z' \cap X_N \neq \emptyset$ and $x_l' \in X_Y \Rightarrow$ the nontrivial SCC that involves (z', θ', x_l') is an NSCC, which contradicts that G_t^a contains no NSCC. $\blacklozenge$